

ISMM C60 Dissertation Project Proposal

Project title: From Asset to Network: Graph-Based Optimisation of Interdependent Maintenance Decisions for Transportation Networks

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Research Centre: Asset Management Group, Institute for Manufacturing

Background

Transport networks are complex, interconnected systems where maintenance decisions made at the asset level can generate cascading effects across an entire region. Traditional maintenance planning typically focuses on individual assets - deciding when and how to intervene on a specific road segment, bridge, or drainage element. While useful for isolated interventions, this asset-level perspective fails to capture interdependencies between projects, which become increasingly important as infrastructures age, budgets tighten, and climate-driven stresses intensify.

Ignoring these interdependencies may lead to suboptimal outcomes: simultaneous works on parallel corridors can cause major congestion; delaying a project may overload alternative routes; and uncoordinated project timing can create network bottlenecks, reduce reliability, and increase lifecycle costs. As transportation agencies shift toward more resilient and data-driven management strategies, there is a growing need for network-level decision-support frameworks that consider how multiple maintenance projects interact and how these interactions influence wider traffic performance, operational disruption, and resource utilisation.

Recent advances in graph-based machine learning, such as Graph Convolutional Networks (GCNs), offer a powerful way to model the topological relationships that define how assets influence each other across space. By combining graph-based network modelling with mathematical optimisation, it becomes possible to evaluate maintenance scenarios holistically, considering both direct impacts on an asset and indirect network-wide effects such as congestion propagation or shifts in route reliability.

This project aims to propose a graph-based optimisation framework for coordinating interdependent maintenance decisions on a road network. By embedding road topology into the optimisation process, the framework enables the evaluation of multiple maintenance projects simultaneously, helping infrastructure managers schedule interventions that minimise cumulative network disruption, reduce long-term costs, and enhance resilience. The approach supports a shift from asset-level decision-making to network-aware maintenance planning, aligning with future demands for adaptable, resilient, and efficient transport systems.

Note: The supervisor, along with Prof. Ajith Kumar Parlikad, is working on a complementary Decarbonised Adaptable and Resilient Transport Infrastructures (DARe) project funded by UKRI and the Department for Transport, a directly related topic to the proposal herein.

Research question

1. How can a graph-based model represent interdependencies between maintenance projects across a transport network?

Objective: Construct a road network graph incorporating topology, traffic flows, and project interactions; develop a graph-based model that captures how conditions or interventions on one link influence neighbouring links and wider corridors.

2. How can optimisation techniques be integrated with graph-based modelling to coordinate multiple interdependent maintenance decisions?

Objective: Develop a portfolio optimisation method that uses graph outputs (e.g., predicted network disruption or load redistribution) to identify maintenance schedules and combinations that minimise cost and network performance degradation.

Suggested methods

The methodology will focus on combining graph-based modelling with mathematical optimisation to support network-aware maintenance planning. The work will be structured into three components:

⑩ Network Graph Construction & Interdependency Modelling [approximately 1 month]:

- 1) Build a graph representation of the road network using available datasets (e.g., link geometry, connectivity, traffic volumes, capacity).
- 2) Identify maintenance projects and map them to corresponding graph nodes/edges.
- 3) Encode spatial, temporal, and functional interdependencies, such as:
 - adjacency (shared nodes or corridors)
 - traffic diversion relationships
 - simultaneous project conflicts
- 4) Train a graph-based model (e.g., Graph Convolutional Network (GCN)) variant to learn:
 - how disruptions propagate
 - how closing or degrading one link affects neighbouring links
 - potential network-level performance consequences

The outcome will be a graph-based model able to estimate network-wide impacts of project combinations.

⑩ Graph-Based Maintenance Portfolio Optimisation [approximately 1 month]:

- 1) Formulate a maintenance scheduling problem as a multi-objective optimisation problem, considering:
 - cost
 - traffic performance degradation
 - network resilience impacts
 - project conflicts and constraints (e.g., resource limits)

- 2) Use optimisation methods such as:
 - Mixed-Integer Linear Programming (MILP)
 - Genetic Algorithms (GA)
 - Heuristic portfolio optimisation
- 3) Integrate graph-derived impact estimates as constraints or objective functions.
- 4) Evaluate trade-offs between alternative scheduling combinations and prioritisation strategies.

The outcome will be an optimisation engine that selects maintenance project portfolios minimising disruption and costs while supporting network resilience.

⑩ Case Study & Decision Support Prototype [approximately 1 month]:

- 1) Apply the integrated graph + optimisation framework on a representative network (e.g., urban network, or a similar dataset that we have access to).
- 2) Test multiple scheduling scenarios (e.g., clustered works, staggered works, capacity-restricted corridors).
- 3) Compare:
 - optimised network-aware scheduling
 - conventional single-project optimisation
- 4) Assess improvements in:
 - cumulative maintenance cost
 - congestion/disruption levels
 - network resilience (e.g., reduced bottlenecks, load balancing)
- 5) Develop a simple decision support prototype to visualise:
 - network impacts
 - recommended schedules
 - trade-off curves

The outcome will be a demonstration of how graph-based optimisation enhances network-level decision-making.

The proposed methodology will be validated through a case study utilizing network-level data comprising records for road network in the U.S. or China.

Recommended bibliography:

1. Kipf, T. N. (2016). Semi-supervised classification with graph convolutional networks. arXiv preprint arXiv:1609.02907.
2. Gao, H., Wang, Z., & Ji, S. (2018, July). Large-scale learnable graph convolutional networks. In Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining (pp. 1416-1424).
3. Liu, T., & Liu, F. (2025). Graph Neural Networks for evaluating the reliability and resilience of infrastructure systems: a systematic review of models, Applications and Future Directions. IEEE Access.

4. Ji, M., Wu, Q., Fan, P., Cheng, N., Chen, W., Wang, J., & Letaief, K. B. (2024). Graph neural networks and deep reinforcement learning based resource allocation for v2x communications. IEEE Internet of Things Journal.

5. Sasidharan, M., Burrow, M. P. N., & Ghataora, G. S. (2020). A whole life cycle approach under uncertainty for economically justifiable ballasted railway track maintenance. Research in Transportation Economics, 80, 100815.

Student skills, abilities or knowledge

Conceptual thinking; systems thinking; machine learning (GCNs preferable but not required); excellent writing skills (since the proposed project has the potential for a publication); coding skills (e.g., python).